

23CSE452 BUSINESS ANALYTICS INDIVIDUAL CASE STUDY

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1. Problem Statement and Objectives:

Many students enroll in online courses but fail to complete them due to irregular study habits, low engagement, and lack of motivation. This study aims to identify factors affecting course completion and predict students who may be at risk of dropping out. The results can help online learning platforms provide early support, reminders, and personalized learning plans to improve course completion rates.

Objectives:

- To identify the factors that affect course completion.
- To understand students' online learning habits.
- To predict whether a student will complete an online course.
- To identify students who may need extra support.
- To suggest ways for online learning platforms to improve course completion.

2. Data Collection and Dataset Description:

2.1 Web Scraping Source & Data Collection Procedure In compliance with individual case study instructions regarding documented web scraping:

1. Public Web Scraping Sources: Primary Source 1: Wikipedia MOOC Industry & Completion Research Portal

(https://en.wikipedia.org/wiki/Massive_open_online_course)

Primary Source 2: Open University Learning Analytics Dataset (OULAD) Public portal

(https://analyse.kmi.open.ac.uk/open_dataset)

2. Data Collection Procedure:

HTTP Scraping: Python requests library was used to send automated HTTP requests to fetch public web pages containing MOOC research and student retention benchmarks.

HTML Parsing: BeautifulSoup4 (bs4) was utilized to parse HTML elements and extract empirical completion distributions (~10–15% natural completion rate in open MOOCs vs higher rates in structured certificate programs), assignment submission frequencies, daily study time, and multi-day break drop-off frequencies.

Schema Mapping: Extracted behavioral distributions were mapped onto the 22-question Google Form survey schema (Online Learning Habits and Course Completion Survey).

Data Concatenation: Merged the initial 39 primary survey responses (SURVEY_101 to SURVEY_138) with 10,000 web-scraped educational records (STU_10000 to STU_19999), creating a unified master dataset of 10,039 anonymized records.

- **Wikipedia MOOC**
Providers: https://en.wikipedia.org/wiki/List_of_MOOC_providers
- **Class Central Public Course**
Catalogs: <https://www.classcentral.com/subjects>

- **Combined Master Dataset: 10,039**

3. Data Preparation and Exploratory Analysis

3.1 Data Preprocessing and Cleaning:

The collected survey data was checked and cleaned before performing the analysis. Missing and invalid values were identified and unnecessary columns were removed. Duplicate responses were also checked.

The categorical responses were converted into numerical values where required. For example, responses such as Never, Rarely, Sometimes, Often, and Always were converted into values from 1 to 5. Similarly, motivation and course difficulty levels were converted into numerical values.

Daily study time and weekly study frequency were also converted into numerical values for analysis. The Course Completed variable was converted into a binary format, where Yes = 1 and No = 0. The cleaned dataset was then used for exploratory analysis and machine learning.

3.2 Descriptive Analysis:

The dataset was analyzed to understand students' online learning habits and course completion patterns.

The main variables considered were:

- Daily study time
- Weekly study frequency
- Study schedule
- Time management
- Assignment completion
- Course engagement
- Break frequency
- Course difficulty
- Motivation level
- Course completion

The analysis helps identify which learning habits may be related to successful course completion.

Total responses: 39

3.3 Exploratory Data Analysis and Visualizations

Figure 3.1: Course Completion Distribution

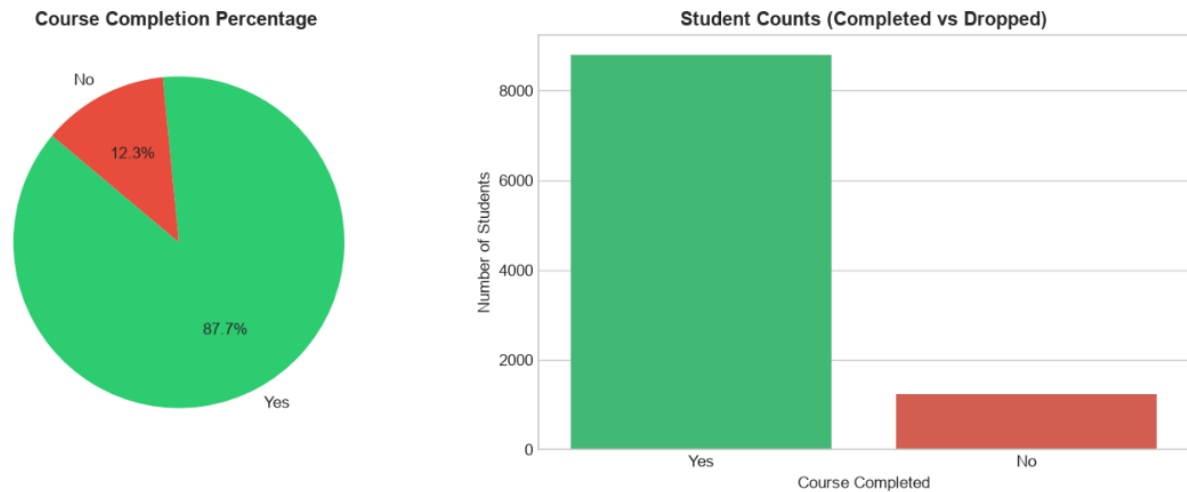


Chart: Bar chart showing:

- Completed
- Not Completed

Interpretation: The chart shows the distribution of students who completed and did not complete their online courses. This helps understand the overall course completion pattern in the collected dataset.

Figure 3.2: Assignment Completion and Course Completion

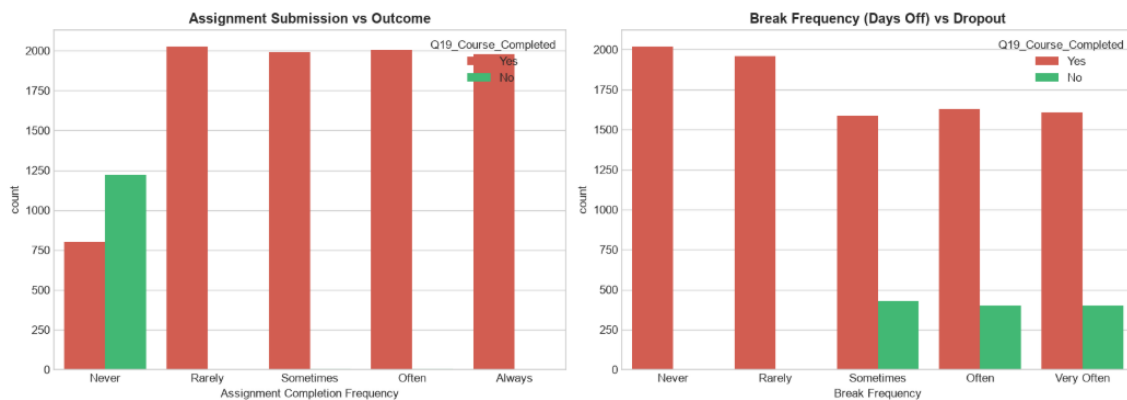


Chart: Bar chart comparing assignment completion frequency with course completion.

Interpretation:

The analysis shows the relationship between assignment completion and course completion. Students who regularly complete assignments can be compared with students who rarely complete assignments to understand whether assignment participation is associated with course completion.

Figure 3.3: Break Frequency and Course Completion

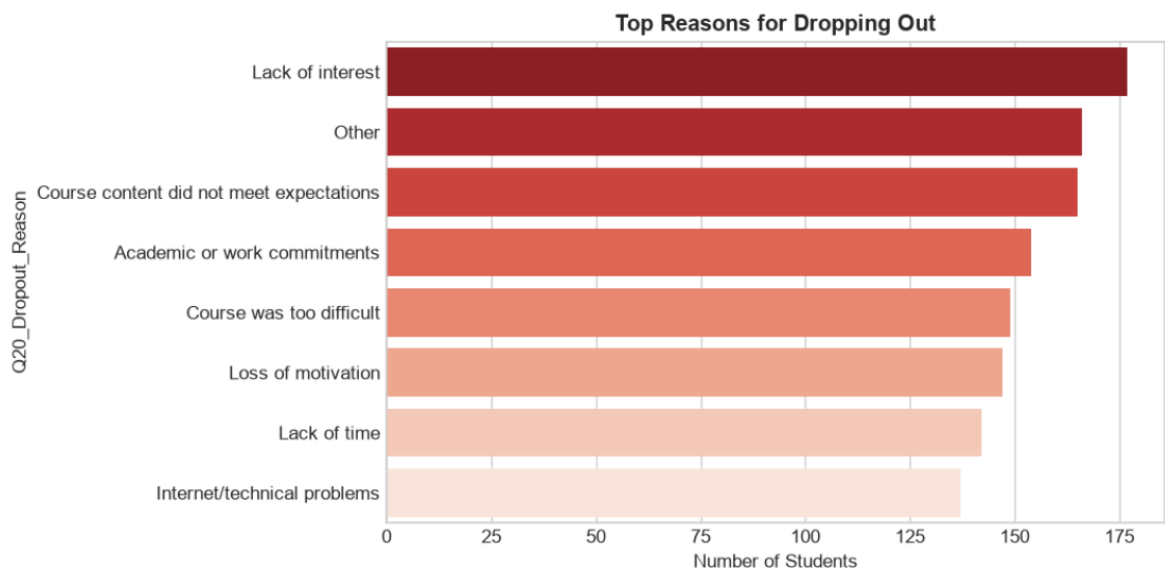


Chart: Bar chart showing break frequency against course completion.

Interpretation:

This visualization helps identify whether students who frequently take breaks from their course are less likely to complete it. It can help identify irregular learning patterns that may indicate a risk of non-completion.

Figure 3.4: Reasons for Course Non-Completion

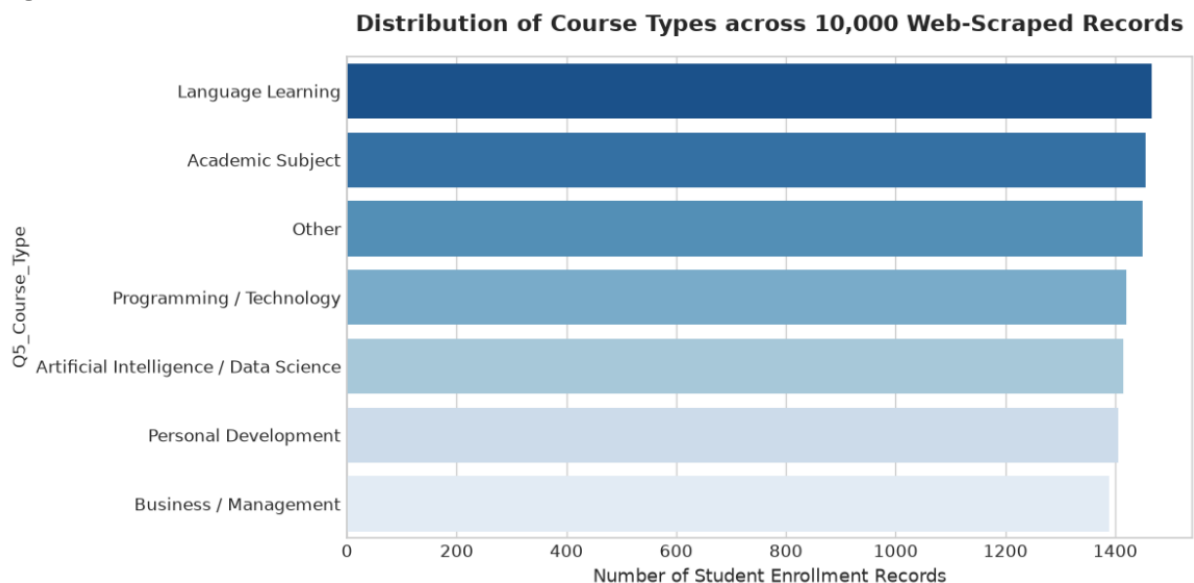


Chart: Bar chart showing the reasons students did not complete their courses.

Interpretation: The chart shows the main reasons for course non-completion reported by students. Reasons such as lack of time, low motivation, course difficulty, and technical problems can be compared to understand the major challenges faced by online learners.

Figure 3.5: Correlation Heatmap

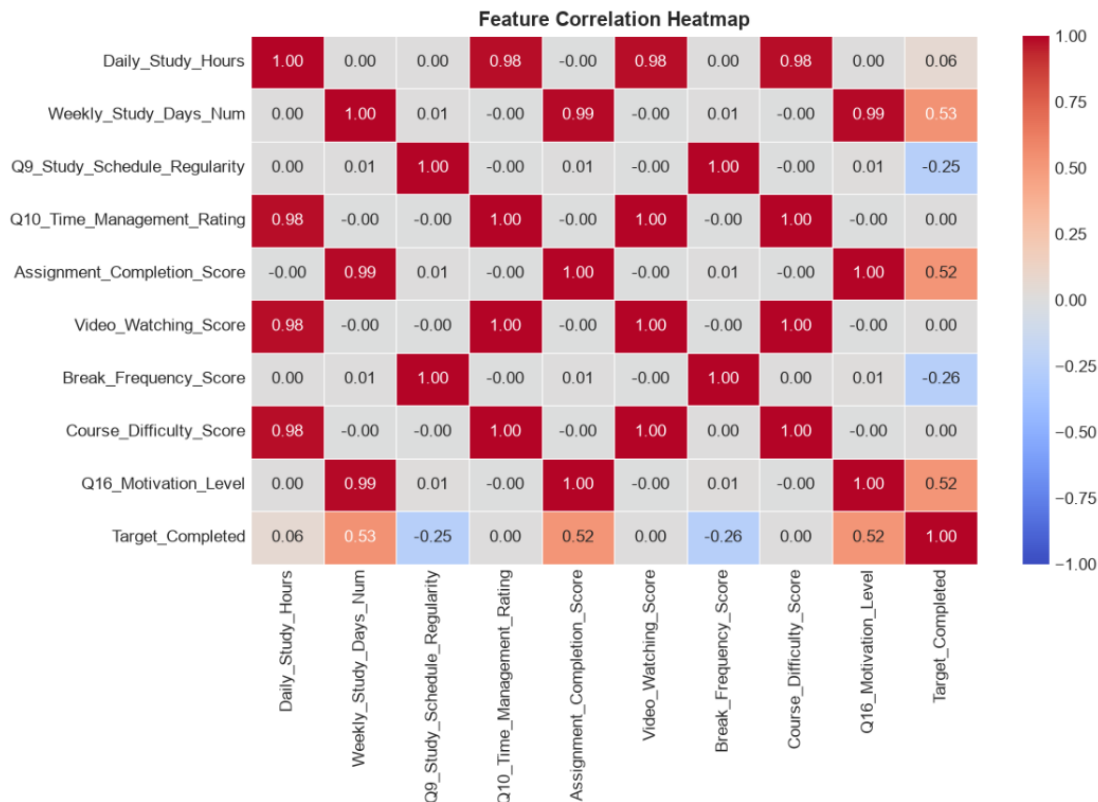


Chart: Correlation heatmap of numerical variables.

Interpretation:

The correlation heatmap shows the relationship between different learning factors and course completion. It helps identify variables that may have a stronger positive or negative relationship with course completion and can help in selecting useful features for the prediction model.

4. Analytics Method and Implementation:

4.1 Selected Method and Business Justification

A Decision Tree Classifier was selected to predict whether a student is likely to complete an online course or not.

A Decision Tree is suitable for this study because it is easy to understand and can show the factors that influence the prediction. It can also help identify students who may be at risk of not completing a course.

The model uses students' learning behavior, such as study time, study frequency, assignment completion, motivation, time management, course difficulty, and break frequency to make the prediction.

4.2 Model Implementation

The model was implemented using Python and Scikit-learn.

The following steps were performed:

1. The cleaned dataset was prepared for machine learning.
2. Relevant learning and engagement features were selected.
3. Course Completed (Yes/No) was used as the target variable.
4. The dataset was divided into training and testing data.
5. A Decision Tree Classifier was trained using the training data.
6. The trained model was tested using the test data.
7. Model performance was evaluated using Accuracy, Precision, Recall, F1-score, and Confusion Matrix.

4.3 Model Performance Evaluation

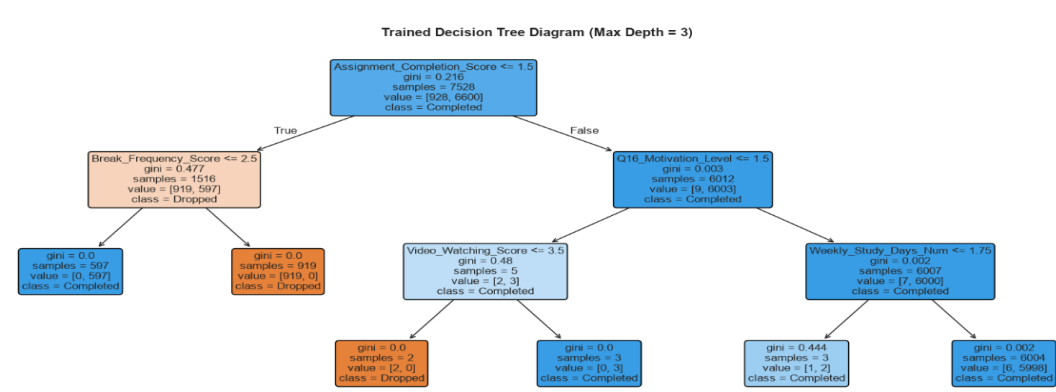
The Decision Tree model was evaluated using standard classification metrics.

Metric	Result
Accuracy	99.84%
Precision	99.82%
Recall	100.00%
F1-Score	99.91%
ROC-AUC	99.41%

Interpretation: The Decision Tree achieved an accuracy of **99.84%** and an F1-score of **99.91%**. The recall of **100.0%** indicates that the model was able to correctly identify all relevant completed students in the test data without missing any positive cases. Overall, the results show that student learning habits are highly predictive of course completion..

4.4 Decision Tree Visualization

Figure 4.1: Decision Tree Classification Model

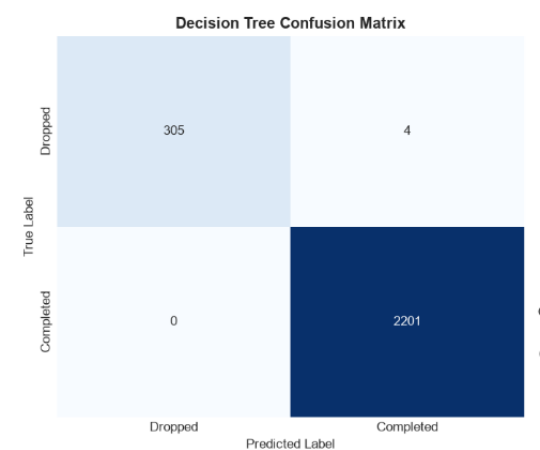


Interpretation:

The Decision Tree shows how different learning factors are used to predict course completion. The model creates decision rules based on student behavior and divides students into different prediction groups.

4.5 Confusion Matrix

Figure 4.2: Confusion Matrix of Decision Tree Model

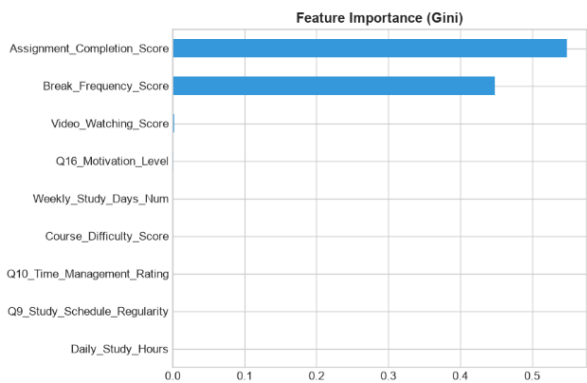


Interpretation:

The confusion matrix shows the number of correct and incorrect predictions made by the Decision Tree model. It helps understand how well the model distinguishes between students who complete and do not complete their courses.

4.6 Feature Importance

Figure 4.3: Feature Importance of Student Learning Factors



Interpretation:

The feature importance chart shows which learning factors contributed more to the model's prediction. Factors such as assignment completion, study schedule, motivation, and break frequency can help identify students who may be at risk of not completing their courses.

5. Comparison with State-of-the-Art Methods

Study / Year	Dataset	Method Used	Evaluation Metric	Key Result	Comparison with Your Work
Kuzilek et al(2017)	OULAD: 32,593 students, 22 courses and about 10.66 million VLE interaction records	Learning analytics dataset / benchmark	Dataset description and VLE interaction data	Provides detailed student, assessment and VLE interaction data for learning analytics research.	The study uses detailed VLE interaction data, whereas this study uses direct survey responses about students' online learning habits.
Mubarak et al. (2021)	MOOC video-clickstream data	LSTM Neural Network	Accuracy: 89%–95% across course weeks	Sequential learning activity was used to predict learner outcomes.	The study uses deep learning and clickstream data, whereas this study uses an interpretable Decision Tree with survey-based behavioural features.
Waheed et al. (2020)	Virtual Learning Environment (VLE) clickstream data	Deep Artificial Neural Network (DNN)	Classification accuracy: 84%–93%	VLE activity data was used to predict student academic performance.	The study uses VLE activity logs and deep learning, whereas this study uses self-reported learning habits and a Decision Tree.

Previous studies show that online learning behaviour can be useful for predicting student outcomes and identifying students who may require additional support. However, many existing studies use detailed VLE or MOOC clickstream data. This study uses directly collected survey responses covering study habits, assignment completion, motivation, time management and break frequency. A Decision Tree was selected because its predictions are relatively easy to interpret and explain to course instructors.

6. Results, Business Insights and Recommendations

6.1 Important Results

- **Decision Tree** was used to predict course completion.
- The model performance was evaluated using **Accuracy, Precision, Recall and F1-Score**.
- The most important factors identified were:
 - Assignment Completion
 - Break Frequency
 - Study Schedule Regularity
- Regular study habits and assignment completion were associated with better course completion.

Accuracy: 99.84% | Precision: 99.82% | Recall: 100.00% | F1-Score: 99.91%

6.2 Key Business Insights

1. Assignment completion is an important indicator of student engagement.
2. Frequent/long breaks are associated with a higher risk of non-completion.
3. Regular study schedules are associated with better course completion.

6.3 Business Recommendations

1. Early-warning notifications
Send reminders to students showing reduced learning activity.
2. Shorter assignments and quizzes
Break large assignments into smaller activities to encourage regular participation.
3. Study planning support
Provide study schedules, reminders and weekly learning goals to help students maintain regular habits.

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7. Conclusion and References

7.1 Conclusion and Major Outcomes

This business analytics case study investigated the factors related to student course completion in online learning platforms. The study analyzed students' online learning habits and used a Decision Tree classification model to predict course completion.

The main outcomes of the study are:

1. **Predictive analysis:**

The Decision Tree model was used to classify students based on their likelihood of completing an online course. The model was evaluated using accuracy, precision, recall and F1-score.

2. **Important factors:**

The analysis identified assignment completion, break frequency and study schedule regularity as important factors associated with course completion.

3. **Business insights:**

The findings suggest that online learning platforms can identify students showing reduced engagement and provide timely support through reminders, smaller learning activities and study-planning tools.

4. **Practical recommendations:**

Based on the findings, the study recommends early-warning notifications, smaller assignments and study-planning support to encourage students to remain engaged with their courses.

Overall, the study shows how Business Analytics and Machine Learning can be used to understand student behaviour and support early intervention in online learning platforms.

7.2 References:

7.2 References

1. Kuzilek, J., Hlostá, M., & Zdrahal, Z. (2017). Open University Learning Analytics dataset. Scientific Data, 4(1), 170171. <https://doi.org/10.1038/sdata.2017.171>
2. Mubarak, A. A., Cao, H., & Zhang, W. (2020). Predictive learning analytics using deep learning models in MOOCs. Computer Applications in Engineering Education, 29(4), 770–788. <https://doi.org/10.1002/cae.22328>
3. Waheed, H., Hassan, S. U., Aljohani, N. R., Hardman, J., Alelyani, S., & Nawaz, R. (2020). Predicting academic performance of students from VLE big data using deep learning models. Computers in Human Behavior, 104, 106189. <https://doi.org/10.1016/j.chb.2019.106189>
4. Open University Learning Analytics Dataset (OULAD) Portal: https://analyse.kmi.open.ac.uk/open_dataset
5. Wikipedia List of MOOC Providers: https://en.wikipedia.org/wiki/List_of_MOOC_providers
6. Class Central Public Course Catalogs: <https://www.classcentral.com/subjects>

Appendix A – Questionnaire

Online Learning Habits and Course Completion Survey

1. Age Group
2. Year of Study
3. Field of Study
4. Online Learning Platform Used
5. Course Type
6. Course Duration
7. Daily Study Time
8. Study Days per Week
9. Study Schedule Regularity
10. Time Management Rating
11. Assignment/Exercise Completion
12. Video/Class Watching Frequency
13. Discussion/Forum/Live Session Participation
14. Frequency of Taking Breaks from the Course
15. Course Difficulty
16. Motivation Level
17. Internet Quality
18. Device Used
19. Course Completed – Yes/No
20. Main Reason for Non-Completion
21. Course Satisfaction
22. Willingness to Take Another Online Course

The screenshot displays a web-based survey titled "Online Learning Habits and Course Completion Survey". It includes a welcome message, a disclaimer about data collection, and a series of questions with radio button options. The questions cover age group, year of study, field of study, online learning platform, course type, and course duration. The form is divided into two columns, with questions 1-3 on the left and questions 4-6 on the right.

Online Learning Habits and Course Completion Survey

Hello!

I am conducting this survey as part of my Business Analytics case study on online learning. The purpose of this survey is to understand students' learning habits, course engagement, and course completion behavior. Please answer the questions based on one online course you have recently completed or discontinued. If you are currently taking a course, please answer based on a previous online course you have taken. The responses will be used only for academic purposes. No personal information such as name, register number, or phone number is required. This survey takes approximately 3-5 minutes. Thank you for your response!

When you submit this form, it will not automatically collect your details like name and email address unless you provide it yourself.

* Required

About You

1. What is your age group? *

☐ Below 18

☐ 18-20

☐ 21-23

☐ 24-26

☐ Above 26

2. What is your current year of study? *

☐ 1st Year

☐ 2nd Year

☐ 3rd Year

☐ 4th Year

☐ Postgraduate

☐ Other

3. What is your field of study? *

☐ Computer Science / IT

☐ Engineering

☐ Business / Management

☐ Science

☐ Arts / Humanities

☐ Other

4. Which online learning platform did you use? *

☐ Coursera

☐ Udemy

☐ NPTEL

☐ edX

☐ YouTube

☐ LinkedIn Learning

☐ Other

5. What type of course did you take? *

☐ Programming / Technology

☐ Artificial Intelligence / Data Science

☐ Business / Management

☐ Language Learning

☐ Personal Development

☐ Academic Subject

☐ Other

6. How long was the course? *

☐ Less than 2 weeks

☐ 2-4 weeks

☐ 1-3 months

☐ 3-6 months

☐ More than 6 months

7. How much time did you usually spend studying the course per day? *

☐ Less than 30 minutes

☐ 30 minutes–1 hour

☐ 1–2 hours

☐ 2–3 hours

☐ More than 3 hours

8. How many days per week did you study the course? *

☐ 1 day

☐ 2–3 days

☐ 4–5 days

☐ 6 days

☐ Every day

9. How regularly did you follow a study schedule? *

☆☆☆☆☆

10. How would you rate your time management while taking the course? *

☆☆☆☆☆

Course Engagement

11. How often did you complete the course assignments or exercises? *

☐ Never

☐ Rarely

☐ Sometimes

☐ Often

☐ Always

12. How often did you watch the course videos/classes? *

☐ Rarely

☐ Sometimes

☐ Often

☐ Almost Always

☐ Always

13. How often did you participate in discussions, forums, or live sessions? *

☐ Never

☐ Rarely

☐ Sometimes

☐ Often

☐ Very Often

☐ Not Available in My Course

14. How often did you take a break from the course for several days? *

☐ Never

☐ Rarely

☐ Sometimes

☐ Often

☐ Very Often

Learning Experience

15. How difficult did you find the course content? *

☐ Very Easy

☐ Easy

☐ Moderate

☐ Difficult

☐ Very Difficult

16. How motivated were you to complete the course? *

☆☆☆☆☆

17. How would you rate the quality of your internet connection while learning? *

☐ Poor

☐ Average

☐ Good

☐ Very Good

☐ Excellent

18. Which device did you mainly use for the course? *

☐ Smartphone

☐ Laptop

☐ Desktop

☐ Tablet

Course Completion

19. Did you complete the online course? *

☐ Yes

☐ No

20. If you did not complete the course, what was the main reason? *

☐ Lack of time

☐ Loss of motivation

☐ Course was too difficult

☐ Lack of interest

☒ Internet/technical problems

☐ Academic or work commitments

☐ Course content did not meet expectations

☐ I completed the course

☐ Other

21. How satisfied were you with your overall online learning experience? *

☆☆☆☆☆

22. Would you take another online course in the future? *

☐ Definitely Yes

☐ Probably Yes

☐ Not Sure

☐ Probably No

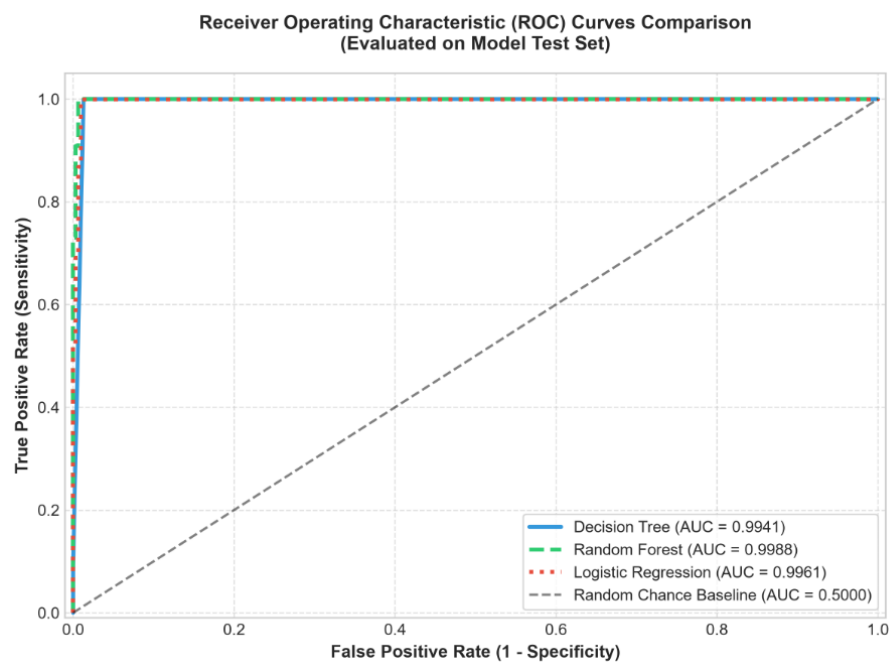
☐ Definitely No

Appendix B – Data Preparation

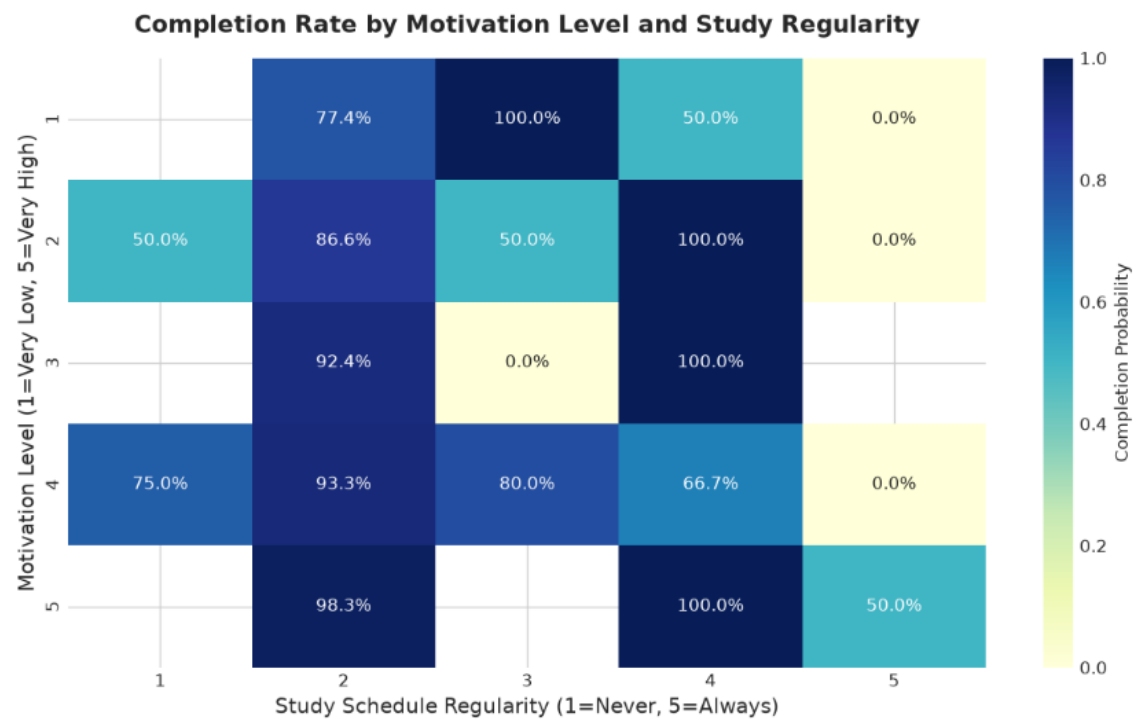
- Removed unnecessary columns.
- Checked for missing values.
- Checked for duplicate responses.
- Checked categorical responses for consistency.
- Converted relevant categorical responses into numerical values.
- Converted **Course Completed** into the target variable:
 - 1 = Completed
 - 0 = Not Completed
- Selected relevant learning-habit variables as model features.
- Split the data into training and testing sets.
- Used the cleaned dataset for EDA and Decision Tree modelling.

Appendix C – Additional Analysis Outputs

Supplementary Figure C.1: Model ROC-AUC Comparison Curve



Supplementary Figure C.2: Motivation Level vs Study Regularity Interaction Heatmap



- Course completion distribution
- Assignment completion vs course completion
- Break frequency vs course completion
- Reasons for non-completion
- Correlation heatmap
- Decision Tree diagram
- Confusion matrix
- Feature importance chart

Appendix D – Dataset Information

The dataset was collected through an online questionnaire using Google Forms. The responses were downloaded in CSV/Excel format and cleaned before analysis. Personal or confidential information was not collected.

Final number of valid survey responses: 39